JONAS MEHTALI

PhD candidate in computer-assisted interventions

jonasx.xyz   jonas.mht.private@gmail.com   j.mehtali@unistra.fr   scholar.google.com/citations?user=0QQM6zIAAAAJ   github.com/JonasMht
linkedin.com/in/jonasmehtali   orcid.org/0009-0003-3688-7782
Strasbourg, France

PROFILE

PhD candidate at ICube (UMR 7357), University of Strasbourg, in the IMAGeS group, in partnership with the Institute of Image-Guided Surgery (IHU Strasbourg), under the supervision of Caroline Essert and Juan Verde. Doctoral training within the ITI HealthTech program, doctoral school MSII.

Research on automatic multi-needle planning for the thermal ablation of liver tumors: trajectory optimization under anatomical constraints, GPU-accelerated bioheat simulation, and planning software designed for clinical use.

EXPERIENCE

PhD researcher *Sep 2024 – present*

ICube (IMAGeS), University of Strasbourg · IHU Strasbourg

- Automatic multi-needle planning for thermal ablation of liver tumors, balancing complete tumor coverage against the risk to vessels and surrounding organs; exhaustive and learning-based planning methods, evaluated in simulation and with clinicians.
- GPU-accelerated multi-resolution bioheat simulation (HEAT) bringing plan evaluation to interactive speeds, and neural cellular automaton estimators (C-NCA) for immediate prediction of ablation zones.
- Intraoperative replanning under respiratory motion, including the design and usability assessment of the planning interface.

End-of-studies internship *Jan – Aug 2024*

ICube (IMAGeS) and IHU Strasbourg

- Real-time thermal ablation simulation on remote GPU compute, with mixed-reality co-planning across devices over a secure connection.
- Design of a custom networking protocol with modified Huffman compression for surgical planning data.

Research internship *Jun – Sep 2023*

TU Darmstadt (MEC-Lab) and ICube (IMAGeS)

- CryoTrack cryoablation planning and navigation: respiratory-gated tracking, electromagnetic localization, and real-time needle guidance.
- Phantom study conducted in the operating rooms of Strasbourg University Hospital; the work led to a MICCAI 2024 paper.

Research internship *May – Jul 2022*

University of Oviedo, Spain

- Full research project, from hypothesis to publication, on the usability of serif and sans-serif typefaces in e-commerce (PeerJ Computer Science).

Research internship *Jun – Jul 2020*

EOST, University of Strasbourg

- Landslide monitoring software for the La Valette site, subsequently taken over and used by EOST students.

PUBLICATIONS AND PATENT

Mehtali, J., Verde, J., Essert, C. **C-NCA: Chained neural cellular automata for fast and accurate thermal ablation estimation.** *MICCAI 2025*, Seoul. doi:10.1007/978-3-032-04965-0_7

Mehtali, J., Verde, J., Essert, C. **HEAT: High-efficiency simulation for thermal ablation therapy.** *International Journal of Computer Assisted Radiology and Surgery*, 2025, 20(6), 1135–1143. doi:10.1007/s11548-025-03350-z

Krumb, H. J., Mehtali, J., Verde, J., Mukhopadhyay, A., Essert, C. **Cryotrack: Planning and navigation for computer assisted cryoablation.** *MICCAI 2024*, Marrakesh. doi:10.1007/978-3-031-72089-5_10

Vecino, S., Mehtali, J., de Andrés, J., Gonzalez-Rodriguez, M., Fernández-Lanvin, D. **How does serif versus sans serif typeface impact the usability of e-commerce websites?** *PeerJ Computer Science*, 2022. doi:10.7717/peerj-cs.1139

Mehtali, J., Essert, C., Verde, J. **Method and apparatus for determining a thermal ablation volume.** European patent application. patents.google.com

EDUCATION

PhD in computer-assisted interventions *2024 – present*
University of Strasbourg, ICube (IMAGeS) · IHU Strasbourg

Master of Science in computer science, image and 3D *2022 – 2024*
University of Strasbourg · M1 12.9/20, M2 13.7/20

Bachelor of Science in computer science, CMI engineering track *2019 – 2022*
University of Strasbourg · 14.3/20

Institut Sainte-Jeanne-d'Arc, Mulhouse *2016 – 2019*
Baccalauréat général 15.4/20 and Allgemeine Hochschulreife (German Abitur) 1.2

SKILLS

Programming	Python, C/C++, CUDA, PyTorch
Medical imaging	3D Slicer, VTK, OpenIGTLink, medical image processing and segmentation
Graphics and mixed reality	OpenGL, Unity, Godot, Blender, Maya, Meta Quest
Engineering	GPU computing, Docker, Linux, Git

LANGUAGES

German (native), French (native), English (C2, LanguageCert Level 3, 2022)

TEACHING AND EARLIER EXPERIENCE

- Teaching in the computer science department at the University of Strasbourg: image processing, medical imaging, 3D graphics, scientific programming in Python and C++.
- Event staff at the Student Life Service (SVU), University of Strasbourg, 2022–2024.
- General duties and shop accounts for the town services of Niffer, Haut-Rhin, 2020–2022.
- IT maintenance intern, Concept Office, Mulhouse, 2018.

Strasbourg, August 2026